Statistics

Chapter 14 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 3

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned}\hat{y} &= 67 + 8x_1 - 10x_2 + 0.004x_3 - 3x_4 \\ x_1 &= \text{number of employees} \\ x_2 &= \text{overhead cost} \\ x_3 &= \text{average markup} \\ x_4 &= \text{theft loss} \\ y &= \text{net profit}\end{aligned}$$

(a) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

We are interested in estimating the profit in this case, which implies that the dependent variable is the net profit.

Step 2

(b) The full name of the given equation is the **multiple regression equation**.

$$\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

Step 3

(c) Evaluate the multiple regression equation at $x_1 = 112$, $x_2 = 65$ (thousand), $x_3 = 50$, and $x_4 = 50$ (thousand).

$$\begin{aligned}\hat{y} &= 67 + 8x_1 - 10x_2 + 0.004x_3 - 3x_4 \\ &= 67 + 8(112) - 10(65) + 0.004(50) - 3(50) \\ &= 163.2\end{aligned}$$

Thus the predicted net sales is 163.2 thousand dollars or $\$163,200$.

Step 4

(d) Given:

$$R^2 \approx 0.86 = 86\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

86% of the variation in the net profit is explained by the variation in the independent variables.

Step 5

(e) Given:

$$s = 3$$

The standard error of the estimate is the average deviation between the actual y-value and the predicted y-value.

Thus the estimated net profit deviates on average by 3 thousand dollars from the actual net profit.

🎯 Result

(a) Net profit

(b) $\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$

(c) 163.2 thousand dollars

(d) 86% of the variation in the net profit is explained by the variation in the independent variables.

(e) The estimated net profit deviates on average by 3 thousand dollars from the actual net profit.

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Solution A

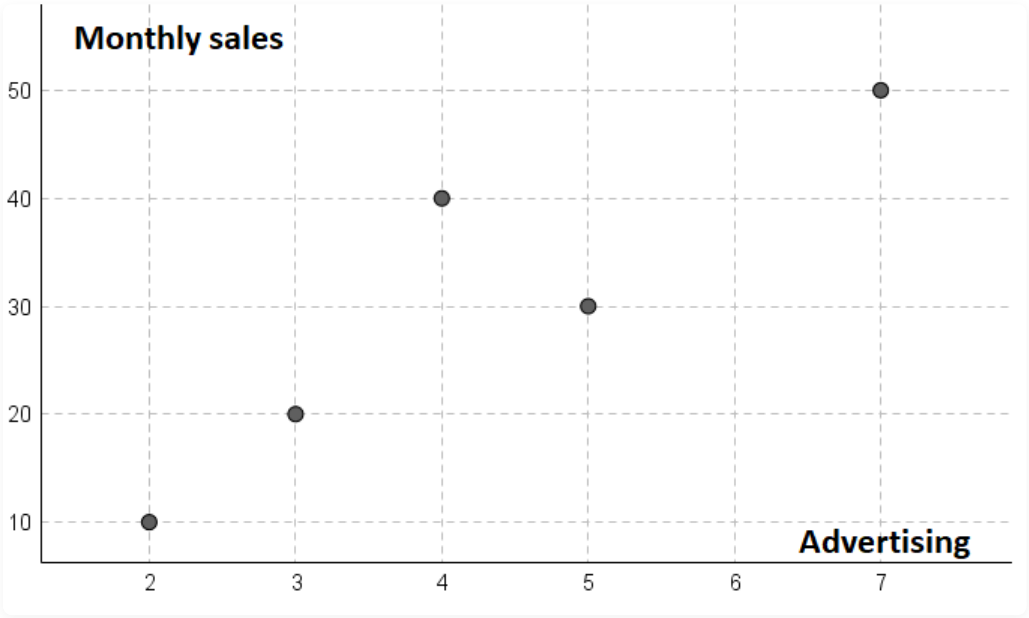
Step 1

Given:

$$n = \text{Sample size} = 5$$

(a) **Scatter diagram**

Annual Bus Bench Advertising is on the horizontal axis and Monthly Sales is on the vertical axis.



Step 2

(b) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2+4+5+7+3}{5} = 4.2$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{10+40+30+50+20}{5} = 30$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2-4.2)^2 + ... + (3-4.2)^2}{5-1}} \approx 1.9235$$
$$s_y = \sqrt{\frac{(10-30)^2 + ... + (20-30)^2}{5-1}} \approx 15.8114$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2	10	-2.2	-20	44
4	40	-0.2	10	-2
5	30	0.8	0	0
7	50	2.8	20	56
3	20	-1.2	-10	12
			Total	110

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{110}{(5-1)(1.9235)(15.8114)} \\ &\approx 0.9042 \end{aligned}$$

Step 4

(c) **Coefficient of determination**

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.9042^2 \approx 0.8176 = 81.76\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

81.76% of the variation in the monthly sales is explained by the variation in the annual bus bench advertising.

Step 5

(d) **Regression equation**

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9042\frac{15.8114}{1.9235} = 7.4324$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 30 - 7.4324 \cdot 4.2 = -1.2162$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -1.2162 + 7.4324x$$

Step 6

(e) Evaluate the least-squares equation found in part (d) at $x = 4.5$ (thousand):

$$\hat{y} = a + bx = -1.2162 + 7.4324(4.5) = 32.2297$$

Thus the estimated monthly sales is 32.2297 thousand dollars or $\$32,229.70$.

Step 7

(f) 81.76% of the variation in the monthly sales is explained by the variation in the annual bus bench advertising.

Since 81.76% is high, the regression equation is appropriate to make predictions.

Result

(a) See Scatter diagram

(b) $r = 0.9042$

(c) 81.76%

(d) $\hat{y} = -1.2162 + 7.4324x$

(e) 32.2297 thousand dollars

(f) Summarize part (a) to (e)

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Solution A

Step 1

(a) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 1050.8$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 1134.6$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{1050.8}{1134.6} \approx 0.9261 = 92.61\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

92.61% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 2

(b) The error sum of squares is given in the row "Error" and in the column "SS" of the ANOVA table:

$$SSE = 83.8$$

The error degrees of freedom is given in the row "Error" and in the column "DF" of the ANOVA table:

$$df_E = 20$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{83.8}{20}} \approx 2.0469$$

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The F- critical value is given in the row with $dfd = df_E = 20$ and in the column with $dfn = df_R = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.87$$

Thus we reject the null hypothesis H_0 when the value of the test statistic is greater than 2.87.

The value of the test statistic is the ratio of the regression mean square and the mean square error, which are the values given in the column "MS" of the ANOVA table.

$$F = \frac{MSR}{MSE} = \frac{262.70}{4.19} \approx 62.6969$$

Since 62.6969 is greater than 2.87, we reject the null hypothesis and thus there is sufficient evidence to support the claim that at least one of the coefficients is nonzero.

Step 4

(d) Determine the critical values from the Student's T distribution table in the appendix in the row with $df = df_E = 20$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.086$$

The rejection region then contains all values smaller than -2.086 and all values greater than 2.086, thus we reject H_0 when $t < -2.086$ or $t > 2.086$.

When the t-ratio is then between -2.086 and 2.086, then we fail to reject the null hypothesis of a zero coefficient and thus the corresponding variable can then be deleted from the model. In this case, we note that variable X_2 has a t-ratio between -2.086 and 2.086, thus variable X_2 can be deleted from the model.

🎯 **Result**

(a) 92.61%

(b) $s = 2.0469$

(c) There is sufficient evidence to support the claim that at least one of the coefficients is nonzero.

(d) Yes